FORM 2

THE PATENTS ACT, 1970
(39 of 1970)
&
The Patents Rules, 2003

COMPLETE SPECIFICATION

(See section 10; rule 13)

1. TITLE OF THE INVENTION

"DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING"

2. APPLICANT (S)

(a) NAME:	NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY
(b) NATIONALITY:	INDIAN INSTITUTE
(c) ADDRESS:	THIRUVETTAKUDY, KARAIKAL - 609 609, PUDUCHERRY, INDIA

3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING

FIELD OF THE INVENTION

The present invention relates to the field of aquaculture sensing systems, and more particularly to a low-cost virtual sensing (VS) system and method for water quality monitoring (WQM). The VS system comprises one or more nodal units, comprising of two or more sensor nodes and a central node, and a water medium. The method of working of the VS system uses less number of sensors, and replaces physical sensor with virtual sensor implemented using solubility equation, inter-dependency relations and long short-term memory (LSTM) in deep learning technique. The present invention provides an affordable and accurate WQM system for all.

BACKGROUND OF THE INVENTION

Aquaculture is vital for global food security as the world population is projected to reach 9.8 billion by 2050. Natural resources are at their limit and the ocean alone may not suffice to meet the demand for food. Aquaculture is an efficient solution for food production, yet its environmental impact needs to be assessed. It is estimated that by 2050, aquaculture will produce one-third of all seafood. Sustainable aquaculture practices are being adopted to ensure the production of healthy and safe food while preserving the environment.

Water quality monitoring (WQM) is an important aspect of aquaculture, as it ensures the health and safety of aquatic organisms. It is essential to track water properties such as temperature, pH, dissolved oxygen (DO), salinity and turbidity to detect changes in water quality. WQM is traditionally done through the use of physical sensors, which are expensive and require frequent maintenance. Thus, there is a need for an alternative and cost-effective sensing system for WQM.

The present invention proposes a low-cost virtual sensing (VS) system and method for WQM. VS is a technique used to obtain readings from water bodies without the use of physical sensors. It is based on the concept of solubility equation and inter-dependency relations. The system comprises one or more nodal units, comprising of two or more sensor nodes and a central node, and a water medium. The method of working of the system uses less number of sensors and replaces physical sensor with virtual sensor implemented using solubility equation and inter-dependency relations. In addition, long short-term memory (LSTM) in deep learning technique is used to improve the accuracy of VS.

Water Quality Monitoring (WQM) is essential for aquaculture to reduce environmental impact, optimize resource utilization and reduce energy wastage. WQM helps to control water pollution, manage water resources, and reduce carbon footprint.

Monitoring fish farms manually is tedious, time-consuming and ineffective, as decisions are only taken after a visible change in the environment has already occurred. Sending water samples to a lab for traditional water quality monitoring also delays decision-making and affects productivity. With the use of sensors to monitor the physical environment, extensive data is provided which may include all the factors that a farmer cannot assess manually. This helps in controlling the system and reducing environmental impact, improving quality of the environment, product quality, and production. WQM helps optimize the use of resources, improving sustainability and allowing the farmer to make a better profit.

The vital water quality parameters in aquaculture are temperature, dissolved oxygen [DO], nitrate, ammonia, turbidity, salinity, pH level and alkalinity. The requirement varies for different species, and farmer needs to be aware of them. The fluctuation in these parameters may result in massive production lose. Several reasons are there for

contamination of water. Even seawater may be polluted due to urbanization and industrialization.

Dissolved Oxygen [DO] is one of the critical water quality parameters, which reflects changes in water quality in Aquaculture. The water quality of an Aquaculture pond has a direct impact on the growth of aquatic animals and product quality. These parameters directly affect fish health and feed utilization affecting the growth rate. Fish experience stress and diseases breakout when the temperature is chronically near their maximum tolerance or fluctuates suddenly.

Water Quality Management (WQM) is an important aspect of aquaculture production, but it is often overlooked by small-scale farmers due to a lack of knowledge and affordability. Many farmers simply assume the water quality is satisfactory and use manual testing kits periodically without proper assessment. Others use chemicals and even evaluate fish behaviour, which can cause stress and affect growth. Most farmers are unaware of technological advances and cannot afford the systems available.

To help farmers become more efficient and sustainable, a low-cost WQM system is needed that is capable of monitoring and passing information to farmers. This system should be modular, portable and affordable, sharing information with farmers in a timely manner, so they can make informed decisions before an outbreak becomes uncontrollable. It should use solubility equations to estimate dissolved oxygen in the water, replacing some of the costly physical sensors.

Reducing the cost of the system will make it more accessible to small-scale farmers, who comprise a significant portion of aquaculture production. With a reliable and affordable WQM system in place, small-scale farmers can manage their farms more effectively, reducing the environmental impacts of aquaculture and increasing production output.

Aquaculture production is increasingly relying on controlled environment systems to maximize yields, requiring regular monitoring of water quality parameters. These parameters can change quickly due to environmental conditions and therefore need to be monitored with high frequency. Existing Water Quality Monitoring (WQM) systems use a set of sensors, gateways, and computers to monitor water quality parameters and water levels, while ZigBee is used for communication. However, these systems are expensive as they require different sensors for each parameter. By sensing each parameter in real-time, the system can reduce the risk of mortality.

This work aims to reduce the cost of water quality monitoring systems utilized in Aquaculture. Aquaculture is a critical factor in providing food security to the world and is becoming more sustainable due to its increasing popularity. Monitoring water quality is essential to improve the sustainability and profitability of aquaculture operations. Parameters such as salinity, temperature, pH, and dissolved oxygen need to be monitored to assess the health of the farm. Moreover, water quality affects fish feeding behavior and growth rate. This research involves the use of cost-effective sensors and replacing costly sensors with Virtual Sensing. This approach can reduce the overall cost of water quality monitoring systems used in Aquaculture and improve the sustainability of operations.

We propose a low-cost water quality monitoring system for aquaculture using virtual sensing. This involves replacing expensive physical dissolved oxygen sensors with solubility equations and inter-dependency relations. Our method leverages the latest advances in open-source hardware and software to reduce the cost of the system. Thus, making it more affordable for small farms. It also helps in identifying reliable sensors with a significant reduction in cost.

This invention (TWI658273B) presents a water quality monitoring system and method. It is composed of at least one sensor device, a water parameters monitoring device, and

a data output display device. The sensor device collects raw data of the water parameters to be measured, while the monitoring device receives the data and uses artificial intelligence, mathematical models, and algorithms to analyze the water parameters and determine the quality of the water. If the water quality is dangerous, the device will issue a warning sound or message and can initiate a water quality improvement system. Additionally, the data is stored in a cloud server for storage and analysis. This system reduces the number of sensors needed to estimate water quality parameters, as well as implements an artificial neural network to estimate values without sensors. Moreover, the microcontrollers hold the algorithm for virtual sensing, which estimates the value of dissolved oxygen and salinity. This system provides monitoring of water quality, storage of data, pre-warning messages, and intelligent starting of the water quality improvement system.

This paper introduces a low-cost, battery-operated and smartphone-based embedded system design to measure different water quality parameters in various remote locations. The system measures pH, total dissolved salt (TDS), and temperature of the water samples with the help of off-the-shelf available sensors. By using standard mathematical relationships such as salinity, oxygen reduction potential, and conductivity, the measured pH and TDS dataset are used to derive other water quality parameters. The front-end readout interface circuit has been designed and interfaced with an 8-bit microcontroller along with a classical Bluetooth module for measurement, data acquisition, and logging purposes.

The dedicated smartphone-based application provides the facility to analyze water quality data with location information on Google Maps for easy understanding. It also offers an auto-calibration feature for on-site usage and sharing of data and warnings using different options such as SMS, WhatsApp, and E-mail. The dimensions of the overall device are 11.0 x 8.0 x 4.0 (in cm), weighs 350 g, and runs with 9- A volt rechargeable battery. The obtained results have been validated with a standard water

quality measurement system from Eutech Instruments, and it was found that measured and calculated parameters are acceptable according to Indian water quality standards.

This invention (CN105549482A) relates to a wireless monitoring system for water environment salinity and dissolved oxygen level. The system includes sensor nodes, a Zigbee wireless transceiving module, a host computer, a liquid crystal display, a power supply module and a solar power supply module. Each sensor node consists of a salinity sensor, a salinity conditioning circuit, a dissolved oxygen amount sensor, a dissolved oxygen amount conditioning circuit and a Zigbee wireless transceiving module, which are all connected. The Zigbee wireless transceiving module is connected with the host computer and the liquid crystal display, and the power supply module supplies power to the Zigbee wireless transceiving module and the liquid crystal display. The solar power supply module supplies power to the sensor nodes. This system is practical and could be used in aquatic product culture production.

This invention (CN101339179B) relates to a water quality remote dynamic monitoring system for aquaculture. It can monitor multiple water parameters such as temperature, pH, electrical conductivity, dissolved oxygen, oxidation-reduction potential, salinity, turbidity and illuminance. It is a wireless monitoring system that uses CDMA communication devices and an information server for radio communication. The system is easy to expand, has higher measurement precision and is compatible with other systems. The cost of this system is more than half of what similar systems imported from abroad cost.

This water quality prediction system (WQPS) and method uses sensors located at the water source and other locations to measure one or more parameters associated with the water source. The parameters are then sent to a fog network computing device which processes them to determine the water quality of the water source. The determined water quality is then displayed on a user device that triggered the request.

This system can be used to measure the water quality of a water source located in different physical locations. The sensors are configured to transmit the parameters to the computing device, which then processes them to determine the water quality and display it on the user device. This system provides a quick and efficient way to measure the water quality of a water source.

Presently available systems use different sensors for sensing the different water quality parametes, making the system expensive. The water quality monitoring has a number of data collection subsystems, which will have all these sensors. These data collection subsystem are deployed in different areas of the water body for collecting data from different parts. The primary cost of the water quality monitoring system is the cost of the sensors. The significant advantages of the proposed scheme, when compared to state of the art, is: this scheme uses the virtual sensing and deep learning (DL) that reduces the cost of the system to an extent. In the present invention, the high cost DO sensor and salinity sensors are replaced with the virtual sensors and uses the deep learning technique to improve its accuracy without compromising on the accuracy of water quality monitoring.

There exists a need for a cost effective virtual sensing system for water quality monitoring (WQM). Further, there exists a need for a method of working of the cost effective virtual sensing system for water quality monitoring (WQM) with deep learning (DL) for accurate water quality monitoring (AWQM).

OBJECTS OF INVENTION

The primary object of the present invention is to provide a low-cost water quality monitoring system that is affordable to small scale farms. At the same time, this system is scalable and can be used in large farms as well.

It is an another object of the present invention to ensure continuous monitoring of the water quality parameters, to ensure a safe environment for fish growth and to optimize the use of the resource.

It is an another object of the present invention to increase the quality of aquaculture production along with an automation in a pond management system.

It is an another object of the present invention, wherein a sustainable solution for aquaculture is proposed.

It is an another object of the present invention, wherein the integration of the present solution with a cloud computing platform significantly enhances the aquaculture industry value chain by monitoring the water quality parameters of the pond, impacting fish health.

It is an another object of the present invention to provide cloud-based data (water quality) monitor, data storage and data process.

It is an another object of the present invention to reduce the manual testing and processing cost.

It is an another object of the present invention to enable the VS system to work using less number of sensors.

It is an another object of the present invention to enable the VS system to use physical sensors replaced by virtual sensors implemented using solubility equation and deep learning (DL).

It is an another object of the present invention to enable the VS system to improve the accuracy of VS using long short-term memory (LSTM) in deep learning technique.

It is an another object of the present invention to provide a low-cost water quality monitoring system which is affordable for all.

SUMMARY OF THE INVENTION

One or more of the problems of the conventional prior art may be overcome by various embodiments of the present invention.

It is a primary aspect of the present invention to provide a virtual sensing system for water quality monitoring (WQM) using solublity equaions and deep learning techniques, comprising:

one or more nodal units that comprises two or more sensor nodes S, and a central node C; and

a water medium,

wherein the central node C is connected to N number of sensor nodes S, each sensor node S comprises of a transceiver, a microcontroller unit and an antenna with two or more sensors, the central node C comprises of a transceiver, a microcontroller unit, a display unit, a Global System for Mobile Communications (GSM) modem, an user device, a server and an antenna,

wherein the sensor node S is deployed in the water medium to acquire water quality parameters, and the estimation of water quality parameter values and environmental parameter values is calculated and the sensed signal is converted to respective water quality parameters by the virtual sensing module and deep learning for error correction in the microcontroller of the sensor node S, the microcontroller of the central node C is provided for controlling and receiving data from the sensor

node S, and the value of water quality parameter values is virtually sensed from acquired electrical signals by the physical sensors and one or more parameters by using the virtually sensing along with deep learning (DL) for error correction/ compensation, and wherein the microcontrollers of the nodal units at the sensor node S is embedded with instructions.

It is an another aspect of the present invention, wherein the sensors acquire environmental parameters such as temperature, pH, conductivity and the like.

It is an another aspect of the present invention, wherein the water quality parameters comprises of dissolved oxygen, salinity and the like.

It is an another aspect of the present invention, wherein the microcontroller unit of the nodal units comprises of a memory module and an input/output module. And the microcontroller unit also has the virtual sensing module and the deep learning (DL) based error correction/compensation unit.

It is an another aspect of the present invention, wherein the memory module is configured to store the water quality parameters and the input/output module is connected to the sensors configured to receive the sensing signals.

It is an another aspect of the present invention, wherein the water quality parameters are sent to the central node C from the sensor node S using the transceiver and the antenna, received at the central node C and stored in a memory; the central node C communicates the information and uploads the water quality parameters to the server. The central node C also uses this data to predict the error in virtual sensing using LSTM deep learning model and is fed back to the Node S for error correction.

It is an another aspect of the present invention, wherein the server is a cloud server.

It is an another aspect of the present invention, wherein the microcontroller is configured to send the water quality parameters information to the user device through the cloud server if online and through the GSM modem if offline.

It is an another aspect of the present invention, wherein the user device comprises of one or more of laptop, mobile, smart devices or any other device.

It is an another aspect of the present invention to provide a method of working of the virtual sensing and deep learning based system for water quality monitoring (WQM), comprising:

initialization of the sensors for virtual sensing of water quality parameters in the water medium and output the sensed signal;

reading the sensors and storing the values in the memory by the microcontroller;

processing the raw data from each sensor and converts to the water quality parameters values by virtual sensing module and LSTM deep learning model based error correction factor from the node C in the microcontroller;

estimation of water quality parameters;

sending data to the central node C;

waiting for predefined time;

receiving data from the sensor node S;

storing the data in the memory;

analyzation of the data;

checking whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated to the user and the data is displayed to the user device by the display unit,

wherein if the parameters are in abnormal range, the information is communicated to the user by an alarm unit,

waiting for predefined time;

wherein the operations are determined to alarm the user or pass the information in a defined time interval, and

displaying the water quality parameters values to the user device by the display unit,

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server. Then the cloud processes all the data received from the sensor node for estimation of the error using the LSTM deep learning model, which is fed back to the node S in each loop helping with error compensation in the virtual sensing in node S.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the features, advantages and objects of the invention, as well as others which will become apparent, may be understood in more detail, more particular description of the invention briefly summarized above may be had by reference to the embodiment thereof which is illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only a preferred embodiment of the invention and is therefore not to be considered limiting of the invention's scope as it may admit to other equally effective embodiments.

Fig. 1 illustrates a block diagram of the cost effective virtual sensing system for water quality monitoring (WQM) with central node and ‘N’ number of sensor nodes according to one embodiment of the present invention.

Fig. 2 illustrates the block diagram of the sensor node of the cost effective virtual sensing and deep learning (DL) based system for water quality monitoring (WQM) according to another embodiment of the present invention.

Fig. 3 illustrates the block diagram of the central node of the cost effective virtual sensing and deep learning (DL) based system for water quality monitoring (WQM) according to another embodiment of the present invention.

Fig. 4 illustrates the flow chart including the method of sensor initialization, data collection, estimation of parameters from sensed values using virtual sensing and error correction factor received from Node C and communicates the same with the Node C in accordance with the present invention.

Fig. 5 illustrates the flow chart of method of receiving the sensed data at Node C, storing data, uploading data to cloud, download error correction factor from cloud, analyzing, determination of the result and communicate the result to the end user in accordance with the present invention.

Fig. 6 illustrates the block diagram of estimation of dissolved oxygen (DO) and creating error dataset from the stored data.

Fig. 7 shows the training and prediction of the error correction fator.

DESCRIPTION FOR DRAWINGS WITH REFERENCE NUMERALS:

[100] Virtual sensing system

[110] Water medium

[120] Central node C

[130a…….130n] Sensor nodes S

[210, 310] Microcontroller unit

[220, 320] Transceiver

[230, 330] Antenna

[240a, 240b, 240c] Sensors

[240d, 240e] Water quality parameters

[340] Display unit

[350] Global System for Mobile Communications (GSM) modem

[360] User device

[370] Server

[400, 500] Method of working of the virtual sensing system for water quality monitoring (WQM)

[410] Initialization of the sensors for sensing water quality parameters in the water medium and output the sensed signal

[420] Reading the sensors and storing the values in the memory by the microcontroller

[430] Processing the raw data from each sensor and converts to the water quality parameters values by the microcontroller

[440] estimation of water quality parameters using solublity equation and deep learning

[450] sending data to the central node C

[460] waiting for predefined time

[510] receiving data from the sensor node S

[520] storing the data in the memory and download deep learning based error correction factor from clould

[530] analyzation of the data

[540] checking whether the parameters are in normal range

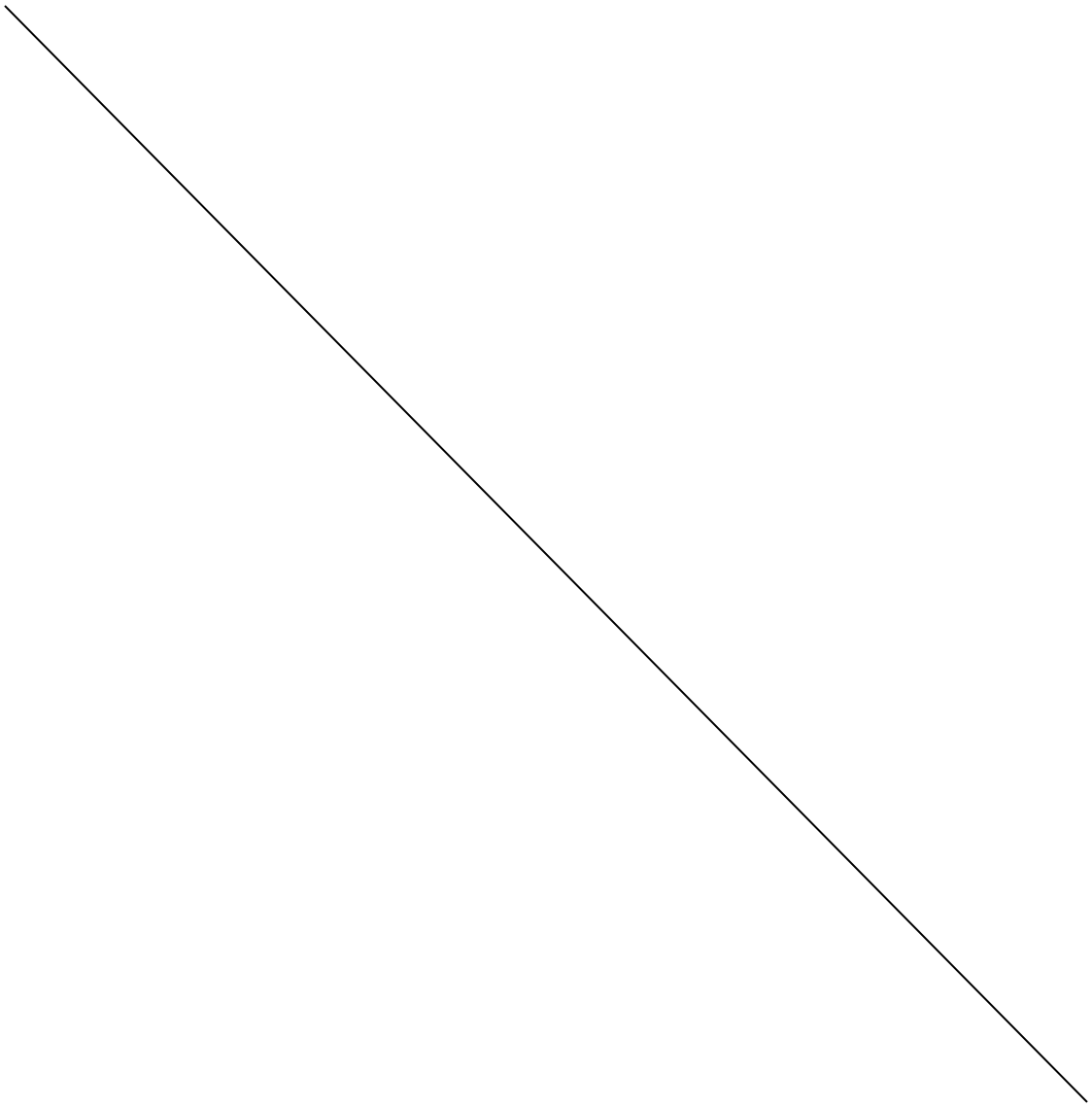
[550] If the parameters are in the normal range, the information is communicated to the user, and the data is displayed to the user device by the display unit

[560] If the parameters are in the abnormal range, the information is communicated to the user by an alarm unit

[570] Waiting for a predefined time

[600] Creating the error dataset from the saved data

[700] Training the LSTM deep learning model using the stored water quality data to predict the error in VS



DETAILED DESCRIPTION OF THE INVENTION

It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The present disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

The present invention relates to the field of aquaculture water sensing systems. More particularly the present invention relates to a cost effective Virtual Sensing (VS) system for water quality monitoring (WQM). Further the present invention relates to a method of working of the cost effective virtual sensing system for water quality monitoring (WQM) using less number of sensors, solubility equation and inter-dependency equations. The present invention uses a long short-term memory (LSTM) based deep learning (DL) model to improve the accuracy of VS.

The novel aspect of this present system is that it uses the virtual sensing (VS) for monitoring water quality parameters and LSTM based deep learning model to replace some physical sensors and to improve the accuracy in monitoring water quality parameters. Replacement of expensive sensors will help in the reduction of the overall cost of the water quality monitoring and reporting system for aquaculture. Aquaculture is an efficient solution for food production but not without impacting the environment. To manage the impact on the environment and improve the efficiency of aquaculture Water Quality Monitoring (WQM) is essential. In this scheme, a novel Virtual Sensing based WQM system for aquaculture is proposed, rather than using separate sensors for each water quality parameters we use virtual sensing to remove some costly sensor reducing the overall cost of the system.

The invention pertains to a low-cost method and an apparatus for monitoring water quality parameters for aquaculture that comprises a central node, a plurality of sensor nodes and an embedded system. The sensors sense water quality parameters and separately output a plurality of sensing signals, including the water quality parameters. The microcontroller system includes a memory module, a virtual sensing module, a deep learning (DL) based error correction module, an input/output module, and a transceiver module. The sensors are connected to the input/output for receiving the sensed signals. The microcontroller converts these sensed electrical signals to corresponding water quality parameters. The memory module stores a plurality of sensor values to the respective water quality parameters. The virtual sensing module in the processor estimates two or more parameters without using the corresponding sensors but uses the sensed parameters; and the error correction in virtual sensing is done using the deep learning (DL) technique. After estimating all the parameters, the data is sent to the central node by the microcontroller. The central node collects this data, stores them in the database, and passes this information to the user in defined intervals. The central node also determines whether the parameters are in abnormal ranges and issues a warning signal according to the determined result. The water quality monitoring system for aquaculture consists of Node C for controlling and receiving the data from many Node S, which have the sensors for sensing water quality parameters. All the water quality data are stored in a cloud server and using the LSTM deep learning model, error in virtual sensing is predicted through cloud/edge computing in the Node C. This error correction factor is downloaded by Node C and then send to Node S. The water quality monitoring system includes a set of sensors and a microcontroller system. The sensors are used to sense the water quality parameters and output a set of sensing signals that corresponds to the water quality parameters. The microcontroller system includes a memory module, a virtual sensing module and an input/output module. The memory module is used to store the water quality parameters. The virtual sensing module estimates the salinity using the signal from the conductivity sensor. And dissolved oxygen (DO) is also estimated by the virtual sensing module from sensed

water quality parameters temperature and estimated salinity. The input/output module connected to the sensors is used to receive the sensing signals. The microcontroller converts all sensed signal to respective water quality parameters and with help of virtual sensing module estimates other parameters without sensors. The water quality parameters are sent to Node C from Node S using the transceiver and antenna, received at Node C and stored in its memory. Further data analysis is done at Node C. Node C communicates the information and uploads the water quality parameters to the cloud.

Fig. 1 shows the water quality monitoring system with the Central Node C [120] and 'N' number of sensor Node S. The sensor Node S [130] may be deployed around the central Node C [120] and may choose any mode of communication with the central Node C. Many sensor Node S [130] is deployed in the waterbody/ water medium [110] to acquire water quality parameters from different areas of the water body/ water medium [110]. The aquaculture water quality monitoring system consists of a central Node C as center point and a set of sensor Node S [130a-n] for sensing water quality parameters and reporting back to central Node C.

The virtual sensing system [100] for water quality monitoring (WQM), comprises one or more nodal units comprises of two or more sensor nodes S [130], and a central node C [120], and a water medium [110]. The central node C [120] is connected to N number of sensor nodes S [130]. Each sensor node S [130a.....130n] comprises of a transceiver [220], a microcontroller unit [210] and an antenna [230] with two or more sensors [240a, 240b, 240c]. The central node C [120] comprises of a transceiver [320], a microcontroller unit [310], a display unit [340], a Global System for Mobile Communications (GSM) modem [350], a user device [360], a server [370] and an antenna [330]. The sensor node S [130] is deployed in the water medium [110] to acquire water quality parameters [240a, 240b, 240c], and the estimation of water quality parameter values [240d, 240e] and environmental parameter values is calculated and the sensed signal is converted to respective water quality parameters

[240d, 240e] by the microcontroller [210] of the sensor node S [130]. The microcontroller [310] of the central node C [120] is provided for controlling and receiving data from the sensor node S [130], and the value of water quality parameter values [240d, 240e] is virtually sensed from acquired electrical signals by the physical sensors [240a, 240b, 240c] and one or more parameters by using the virtually sensed parameters [240d, 240e] and deep learning based error correction for improving the accuracy of VS. The microcontrollers [210, 310] of the nodal units [130, 120] at the sensor node S is embedded with instructions. The sensors acquire environmental parameters such as temperature [240a], pH [240b], conductivity [240c] and the like. The water quality parameters comprises of dissolved oxygen [240d], salinity [240e] and the like. The microcontroller unit [210, 310] of the nodal units [130, 120] comprises of a memory module and an input/output module. The microcontroller unit [210] also have an additional virtual sensing module and deep learning based error correction module which is updated by Node C. The memory module is configured to store the water quality parameters [240a, 240b, 240c, 240d, 240e], the input/output module is connected to the sensors [240a, 240b, 240c] configured to receive the sensing signals, and the virtual sensing unit estimates the water quality parameters without using physical sensors. The water quality parameters [240a, 240b, 240c, 240d, 240e] are sent to the central node C [120] from the sensor node S [130] using the transceiver [220, 320] and the antenna [230, 330], received at the central node C [120] and stored in a memory and the central node C [120] communicates the information and uploads the water quality parameters [240a, 240b, 240c, 240d, 240e] to the server [310]. The server [310] is a cloud server. The microcontroller is configured to send the water quality parameters [240a, 240b, 240c, 240d, 240e] information to the user device [360] through the cloud server [310] if online and through the GSM modem [350] if offline. The user device [360] comprises of one or more of laptop, mobile, smart devices or any other device.

Fig. 2 Shows various parts of central Node S [120] that is used in the water quality monitoring system [100]. The central Node S [120] includes a microcontroller [210], a transceiver [220], and an antenna [230] with the set of sensors [240a-240c]. Sensors [240a – 240c] acquire environmental parameters such as temperature [240a], pH [240b] and conductivity [240c]. The sensor Node S also produces two more water quality parameters without using sensors. The water quality parameters dissolved oxygen [240d] and salinity [240e] is virtually sensed using the parameters already acquired [240a-240c] and using LSTM based deep leaning model for improving the accuracy of VS. The microcontroller [210] includes a memory module, virtual sensing module, deep learning based error correction module, and an input/output module. The input/output module is connected to sensors [240a-240c] and is configured to receive a plurality of sensing signals that correspond to the water quality parameters. The microcontroller processes the signal received from sensors through the input/output module and converts the available raw signals into water quality parameters values with the help of virtual sensing module. Estimation of parameters like DO and salinity is done from temperature and conductivity values using Weiss equation and interdependency equations by the processing of virtual sensing module and deep learning basead error correction module in the microcontroller. All the calculations on the raw sensor data are done in the microcontroller and the processed data is passed to the Node C as water quality parameters.

Fig. 3 illustrates various components in central Node C [130] within the water quality monitoring system [100]. The central Node C [130] includes a microcontroller [310], a transceiver [320] and an antenna [330]. The transceiver receives the data send by sensor Node S through the antenna. The microcontroller depending on the parameter value warns the user and displays the water quality parameters in the LCD [340]. The end-user has the provision to monitor the farm using smartphone [360] by accessing the cloud [370] or communicating with the microcontroller. The LSTM based DL model is implemented in the cloud to estimate the error correction factor. This

estimated error correction factor is sent to the Node S via Node C as Node C only have access to the cloud. The microcontroller has the provision to upload the water quality parameters to the cloud and inform the end through the user smartphone. The microcontroller may also use the GSM modem [350] to send information to the smartphone without the use of cloud as well.

Fig. 4 illustrates a flowchart to initialize and sense the water quality parameters in the water body using the set of sensors and output the sensed signal (410). The microcontroller reads the sensors and store the values in the memory (420). The microcontroller processes the raw data from each sensor and converts to water quality parameters values (430). The processing for estimation (440) of water quality parameters dissolved oxygen and salinity are done from other parameters temperature, conductivity values using equations (1-4) by the microcontroller. The equation 4 is used for processing the estimation of salinity from conductance by the microcontroller. Then the error correction factor is used to improve the accuracy of virtual sensing.

$$S = [5.572*10^{4}\,[SC] + 2.03*10^{-9}\,[SC]^{2}]\,/1000 \quad [1]$$

where S is the salinity in ppt, and T is the water temperature in Kelvin.

Equation 2 is used for processing the estimation of dissolved oxygen from temperature by the microcontroller [210].

$$DO = DO_0 * F_s \quad [2]$$

Where DO is the dissolved oxygen in the water body [110], DO_0 is the Dissolved oxygen concentration at zero salinity and one-atmosphere pressure in the water body [110] given by equation 3. F_S is the salinity factor given in equation 4.

$$DO_0 = 1.42905\, exp[-173.4292 + 249.6339\,[\frac{100}{T}] + 143.3483\,[In\frac{T}{100}] - 21.8492[\frac{T}{100}]] \quad [3]$$

where T is the water temperature in Kelvin and DO is the dissolved oxygen in water mg/l.

$$F_s = exp\left[S\left[-0.033096 + 0.14259\left(\frac{T}{100}\right) - 0.0017\left(\frac{T}{100}\right)^2\right]\right] \quad [4]$$

Where S is the salinity in ppt, and T is the water temperature in Kelvin.

The estimation based on equations is done in Node S and the data is sent to Node C as a water quality parameter. The microcontroller waits for the defined time to initialize the sensors and repeat the process. This scheduling mechanism done in sensor Node S [120] helps the system to be energy efficient.

Fig. 5 illustrates the flowchart to receive the data [510] from Node S, this data is stored in the database [520] and the data is analyzed [530]. The operations [540] are to determine to alarm the user or pass information in a defined time interval [560]. The system waits for a specified time in case of parameters are in the normal range [540]. And the process is repeated continuously passing the information to the end-user and uploading the data to the cloud [560].

The method [400, 500] of working of the virtual sensing system [100] for water quality monitoring (WQM), comprising the following steps: initialization [410] of the sensors [240a, 240b, 240c] for sensing water quality parameters, and virtual sensing of water quality parameters [240d, 240e] in the water medium [110] and output the sensed signal; reading [420] the sensors [240a, 240b, 240c] and storing the values in the memory by the microcontroller [210]; processing [430] the raw data from each sensor and converts to the water quality parameters values [240d, 240e] by the virtual sensing

module in microcontroller [210]; estimation [440] of water quality parameters [240d, 240e]; sending [450] data to the central node C [120]; waiting [460] for predefined time; receiving [510] data from the sensor node S [130]; storing [520] the data in the memory; analyzation [530] of the data; checking [540] whether the parameters are in normal range; if the parameters are in normal range, the information is communicated [550] to the user and the data is displayed to the user device [360] by the display unit [340], if the parameters are in abnormal range, the information is communicated [560] to the user by an alarm unit, waiting [570] for predefined time; the operations [540] is to determine to alarm the user or pass information in a defined time interval [560], and displaying the water quality parameters values [240d, 240e] to the user device [360] by the display unit [340]. The process is repeated continuously passing the information to the end-user and uploading the data to the cloud server [370].

Although, the invention has been described and illustrated with respect to the exemplary embodiments thereof, it should be understood by those skilled in the art that the foregoing and various other changes, omissions and additions may be made therein and thereto, without parting from the spirit and scope of the present invention.

Fig. 6 shows the block diagram of estimation of dissolved oxygen (DO) and creating error dataset from the stored data [600].

Deep learning is used for improving accuracy of VS. LSTM based DL model is used to predict the Error correction factor (ϕ) for reducing the error in virtual sensing. Here for prediction of the error first the error database is created from the monitored data. Error Dataset (DOer)is created for estimating the error correction factor (ϕ DO) using LSTM DL model as in equation 5.

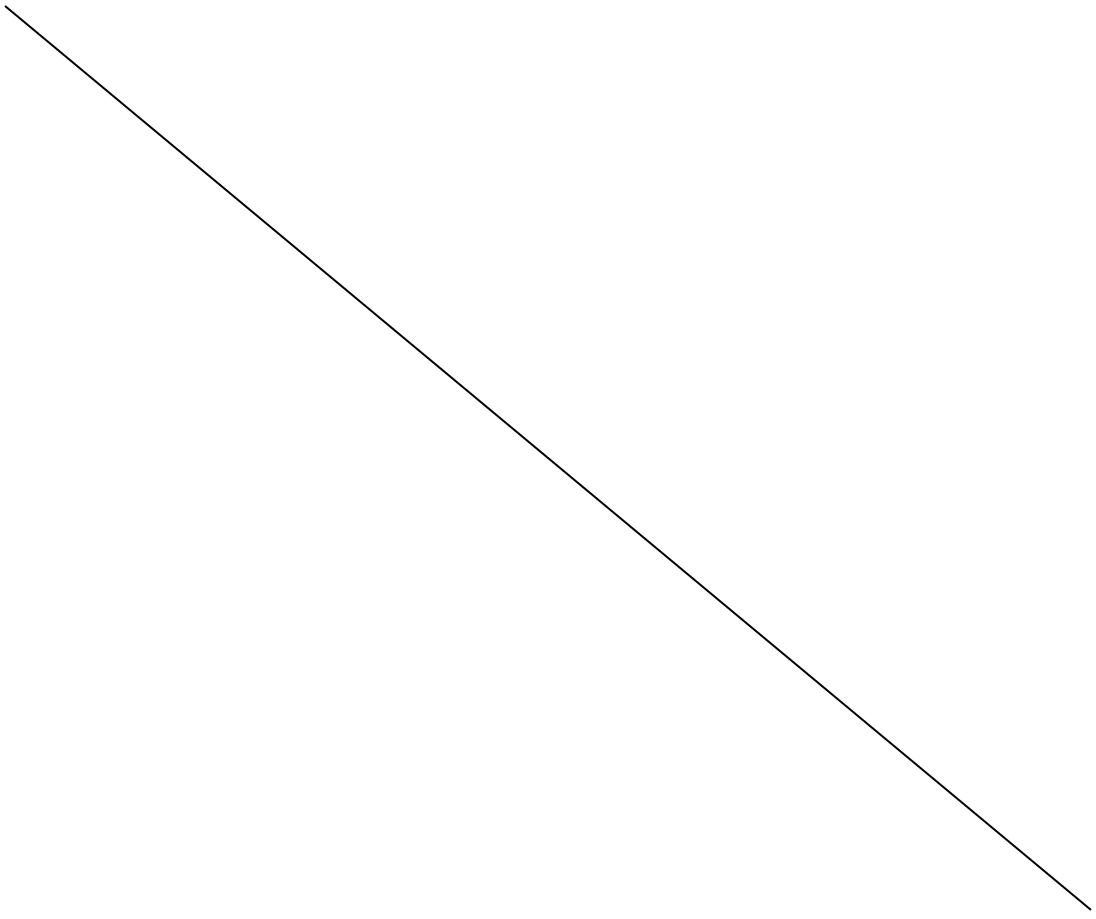
$$DO(t)er = DO(t)vs - DO(t) \quad [5]$$

wherein DO(t)vs is the estimated DO using solubility equations with temperature and salinity data at time t. DO(t) is the original value of DO in the data which is measured using the DO sensor. DO(t)er is the estimated error at time t. t is the time step.

Fig. 7 shows the training and prediction of the error correction fator. The error correction factor ($\phi_{DO)}$) is estimated using the LSTM DL model to reduce error in the virtual sensing [700].

$$DO_R = DO_{vs} - \phi_{DO}$$

Error correction factor (ϕ_{DO}) improve the precision of the virtual sensed DO_{vs} to give DO_R. Here DO_R is the DO which is computed finally after using LSTM to reduce the error of the virtual sensing done by using solubility equation.



NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

WE CLAIM:

1. A virtual sensing system [100] for water quality monitoring (WQM), comprising:

one or more nodal units comprises of two or more sensor nodes S [130], and a central node C [120]; and

a water medium [110],

wherein the central node C [120] is connected to N number of sensor nodes S [130], each sensor node S [130a.....130n] comprises of a transceiver [220], a microcontroller unit [210] and an antenna [230] with two or more sensors [240a, 240b, 240c], the central node C [120] comprises of a transceiver [320], a microcontroller unit [310], a display unit [340], a Global System for Mobile Communications (GSM) modem [350], an user device [360], a server [370] and an antenna [330],

wherein the sensor node S [130] is deployed in the water medium [110] to acquire water quality parameters [240a, 240b, 240c], and the virtual sensing of water quality parameter values [240d, 240e], and environmental parameter values are calculated and the sensed signals are converted to respective water quality parameters [240d, 240e] by the virtual sensing module and deep learning error correction module in the microcontroller [210] of the sensor node S [130], the microcontroller [310] of the central node C [120] is provided for controlling and receiving data from the sensor node S [130], and the value of water quality parameter values [240d, 240e] are virtually sensed from acquired electrical signals by the physical sensors [240a, 240b, 240c] and one or more parameters by using the virtually sensed parameters [240d, 240e], and

wherein the microcontrollers [210, 310] of the nodal units [130, 120] at the sensor node S is embedded with instructions.

2. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the sensors acquire environmental parameters such as temperature sensor [240a], pH sensor [240b], conductivity sensor [240c] and the like.

3. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the water quality parameters comprises of dissolved oxygen sensor [240d], salinity sensor [240e] and the like.

4. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller unit [210, 310] of the nodal units [130, 120] comprises of a memory module, an input/output moduleand a virtual sensing (VS) module.

5. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the memory module is configured to store the water quality parameters [240a, 240b, 240c, 240d, 240e], and the input/output module is connected to the sensors [240a, 240b, 240c] configured to receive the sensing signals.

6. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the water quality parameters [240a, 240b, 240c, 240d, 240e] are sent to the central node C [120] from the sensor node S [130] using the transceiver [220, 320] and the antenna [230, 330], received at the central node C [120] and stored in a memory and the central node C [120] communicates the information and uploads the water quality parameters [240d, 240e] to the server [310].

7. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the server [310] is a cloud server.

8. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller is configured to send the water quality parameters [240a, 240b, 240c, 240d, 240e] information to the user device [360] through the cloud server [310] if online and through the GSM modem [350] if offline.

9. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the user device [360] comprises of one or more of laptop, mobile, smart devices or any other device.

10. A method [400, 500] of working of the virtual sensing system [100] for water quality monitoring (WQM), comprising:

initialization [410] of the sensors [240a, 240b, 240c] for virtual sensing water quality parameters [240d, 240e] in the water medium [110] and output the sensed signal;

reading [420] the sensors [240a, 240b, 240c] and storing the values in the memory by the microcontroller [210];

processing [430] the raw data from each sensor and converts to the water quality parameters values [240d, 240e] by virtual sensing module and deep learning based error correction module of the microcontroller [210];

estimation [440] of water quality parameters [240d, 240e];

sending [450] data to the central node C [120];

waiting [460] for predefined time;

receiving [510] data from the sensor node S [130];

storing [520] the data in the memory and download deep learning based error correction factor;

analyzation [530] of the data;

checking [540] whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated [550] to the user and the data is displayed to the user device [360] by the display unit [340],

wherein if the parameters are in abnormal range, the information is communicated [560] to the user by an alarm unit,

waiting [570] for predefined time;

wherein the operations [540] are determined to alarm the user or pass information in a defined time interval [560], and

displaying the water quality parameters values [240d, 240e] to the user device [360] by the display unit [340],

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server [370], and the cloud server [370] processes all the data received from the sensor node S for estimation of the error using the LSTM deep learning model, which is fed back to the node S in each loop helping with error compensation in the virtual sensing in node S

Dated this 08th day of March 2023



Signature:
Name: Dr. Sudhir Raja Ravindran [IN/PA – 384]
Agent for Applicant

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING

ABSTRACT

The present invention relates to a cost-effective virtual sensing (VS) system for water quality monitoring (WQM). The system comprises one or more nodal units with two or more sensor nodes, a central node, and a water medium. Physical sensors are replaced by virtual sensors based on solubility equations and inter-dependency relations. Moreover, deep learning technique Long Short-Term Memory (LSTM) is used to improve the accuracy of the VS. This low-cost system is affordable for all and has great advantages compared to conventional WQM systems, while requiring fewer sensors. The method of working of the system uses a combination of solubility equations, inter-dependency relations, and LSTM to accurately monitor water quality. This cost-effective system is ideal for aquaculture sensing and provides a reliable and low-cost solution for water quality monitoring.

Reference Figure 4, 5 and 6